

Generalizable Dynamics Models for Deformable Objects: Tool-Agnostic Model for Tool Geometry Design

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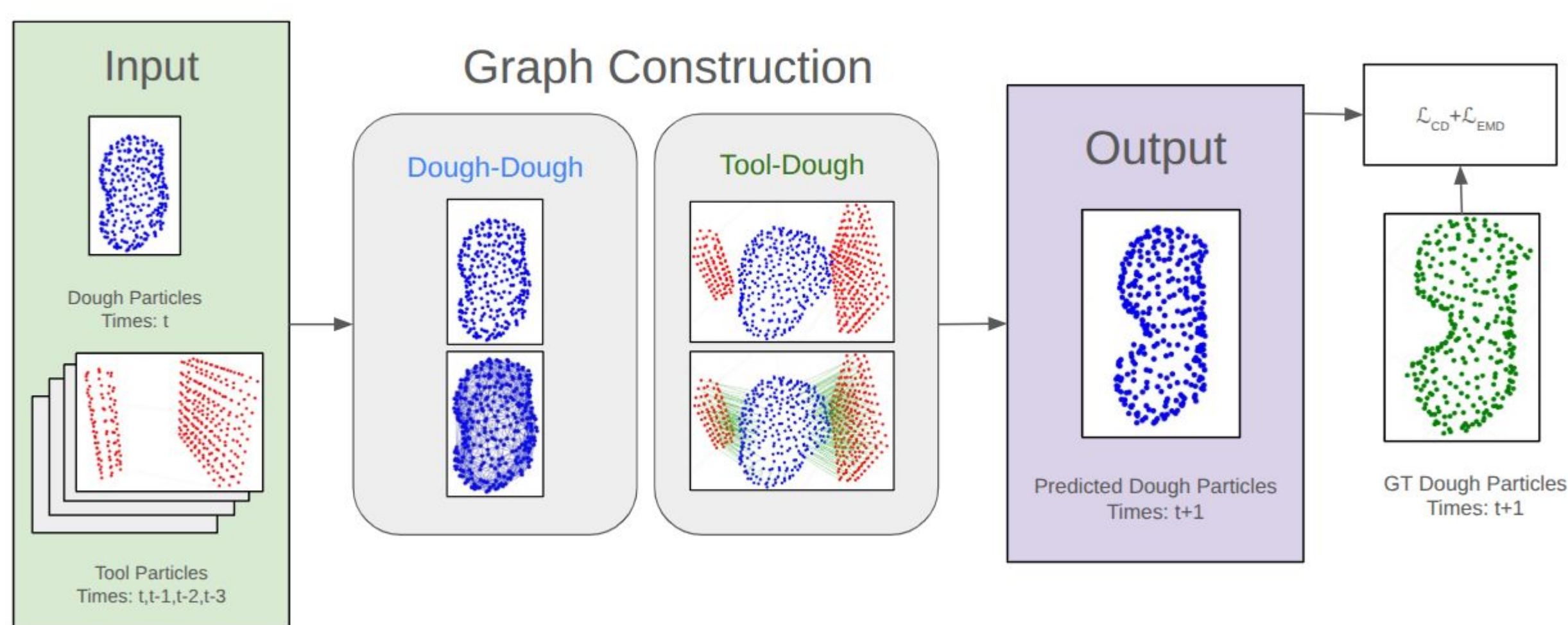
Project Page:



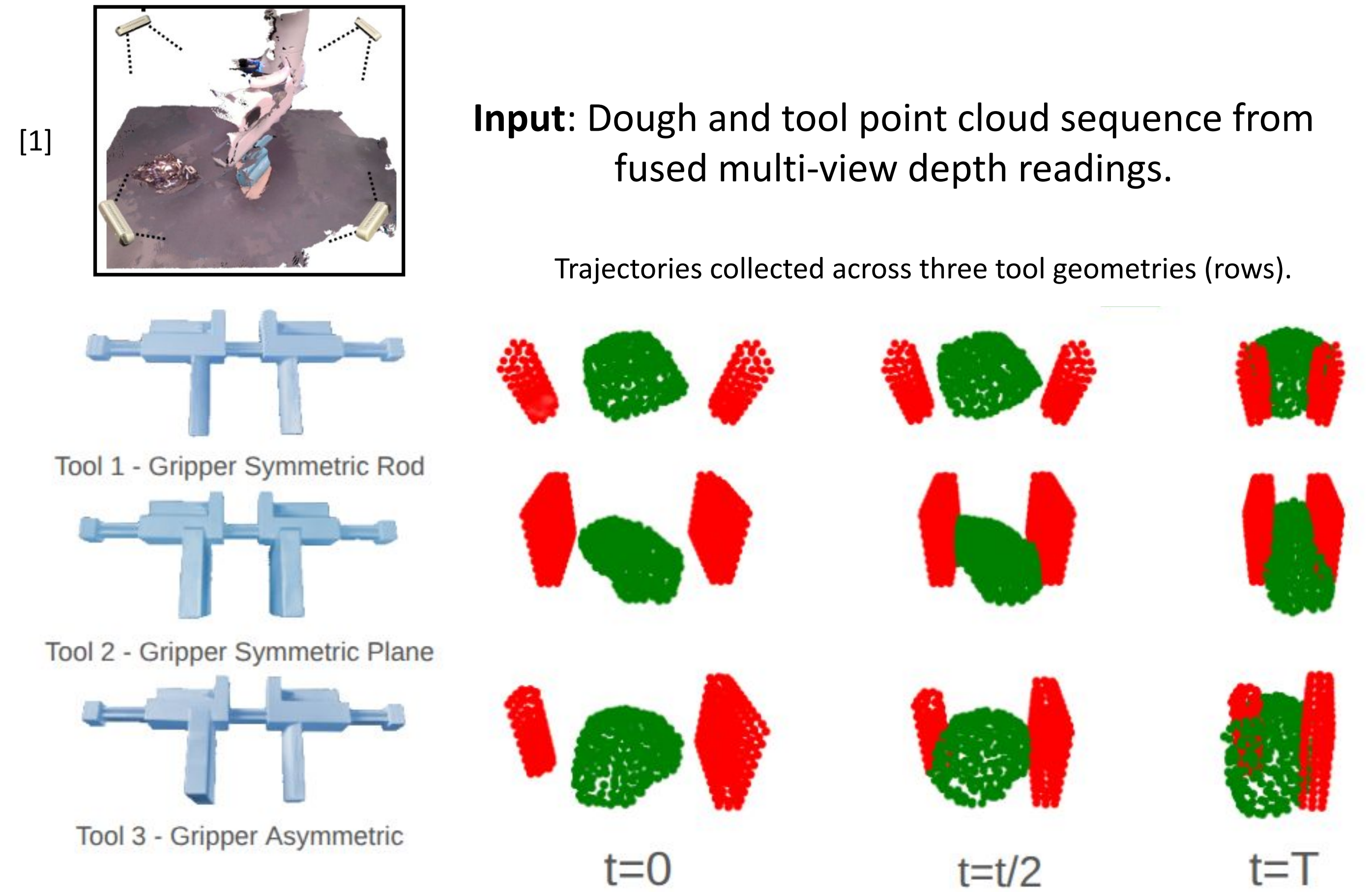
Motivation

- Model-based planning needs dynamics that stay **accurate across many scene configurations**.
- For elastoplastic objects (dough), **analytic models** are too inaccurate.
- Learned models hallucinate** outside their training distribution.
- Prior tool-design methods use explicit templates / cages, **limited shape expressivity** due to low-dimensional representation.

GNN Dynamics Model



Model Training



Results

Quantitative Evaluation: In-distribution

# Train tools	CD↓	EMD↓	HD↓	SDF↓
1 tool [1]	6.08	4.65	16.77	32.9
2 tools	5.73	4.02	15.24	29.28
3 tools	5.74	4.20	14.36	22.76

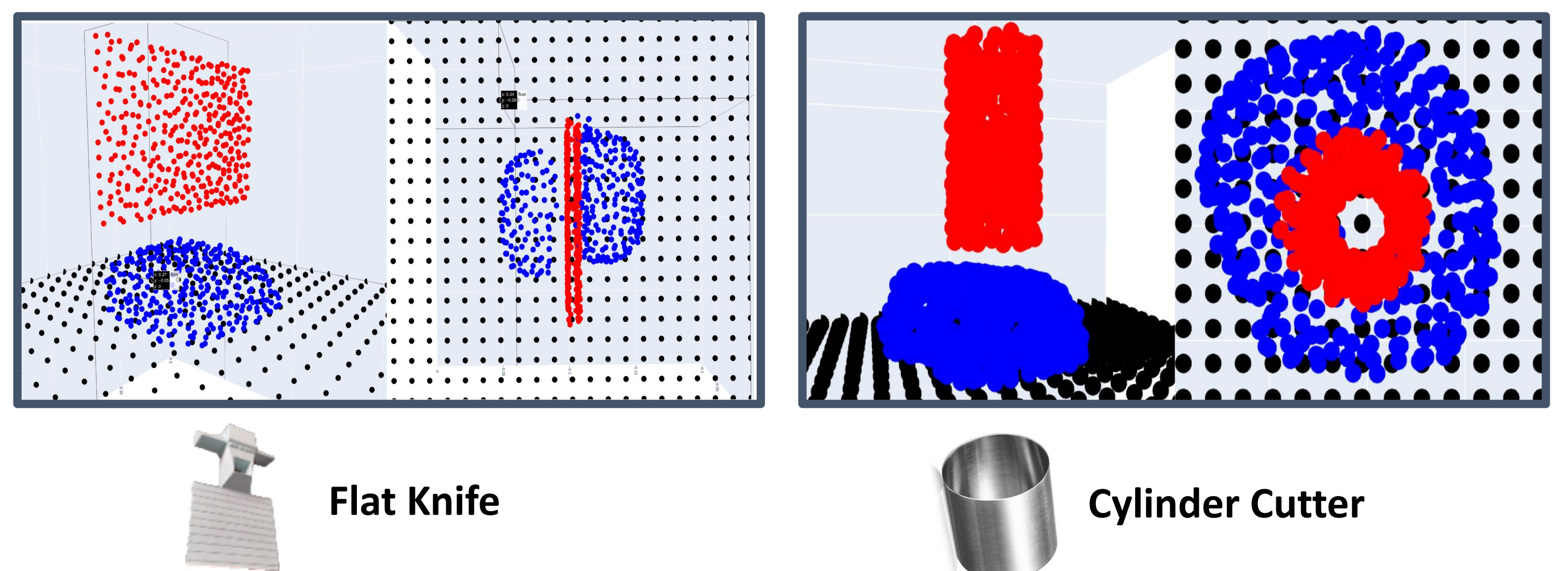
Tested on a held-in tool (in mm). Metrics: Chamfer (CD), Earth Mover's (EMD), Hausdorff (HD) distance, and signed-distance penetration (SDF).

Quantitative Evaluation: Out-of-distribution

# Train tools	CD↓	EMD↓	HD↓	SDF↓
1 tool	7.13	6.39	23.5	20.72
2 tools	7.25	6.37	20.51	17.46

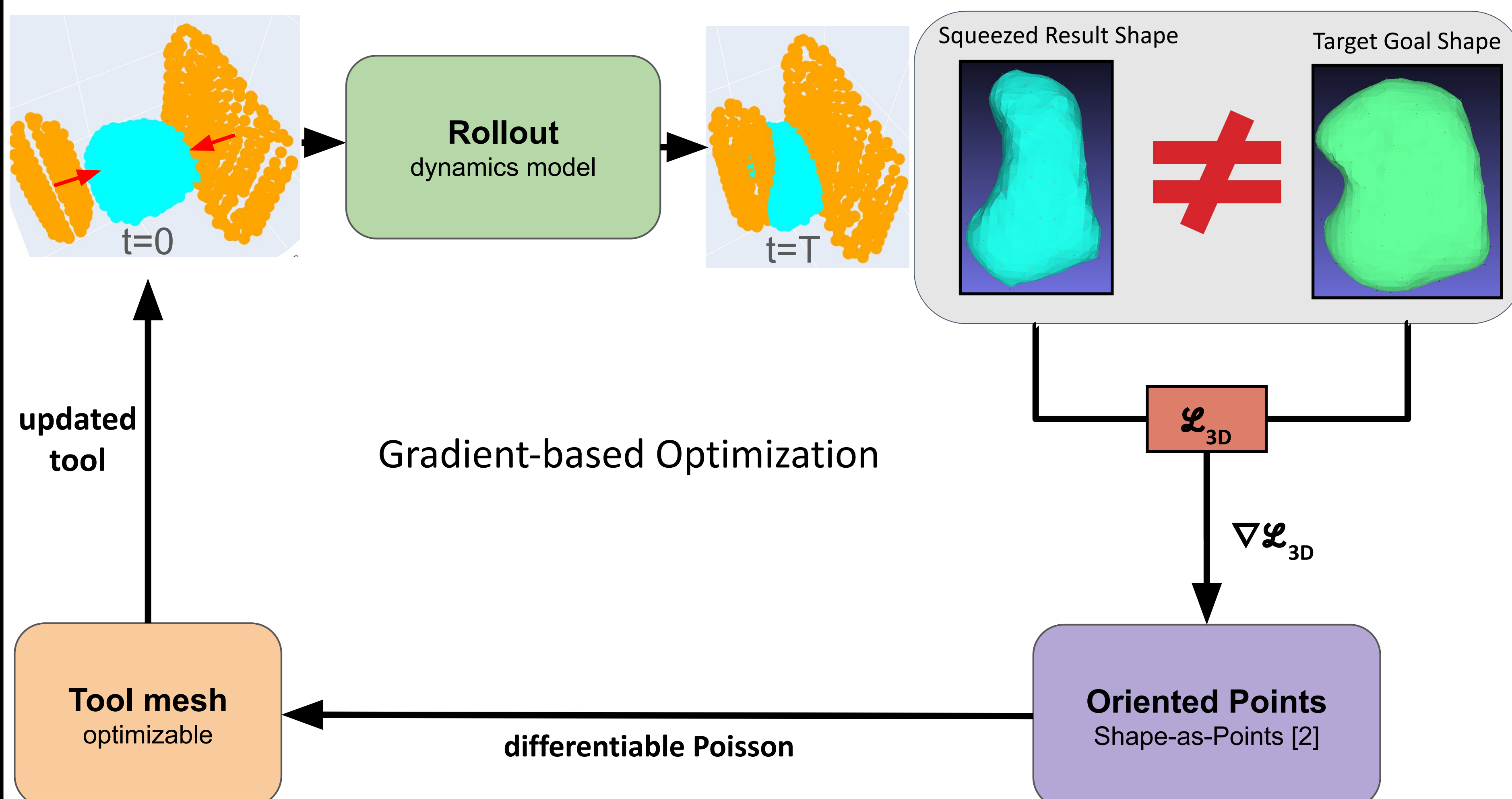
Qualitative Evaluation (OOD Tools)

Trained on push & pinch tools, model is evaluated on these unseen tools.



Tool Shape Optimization

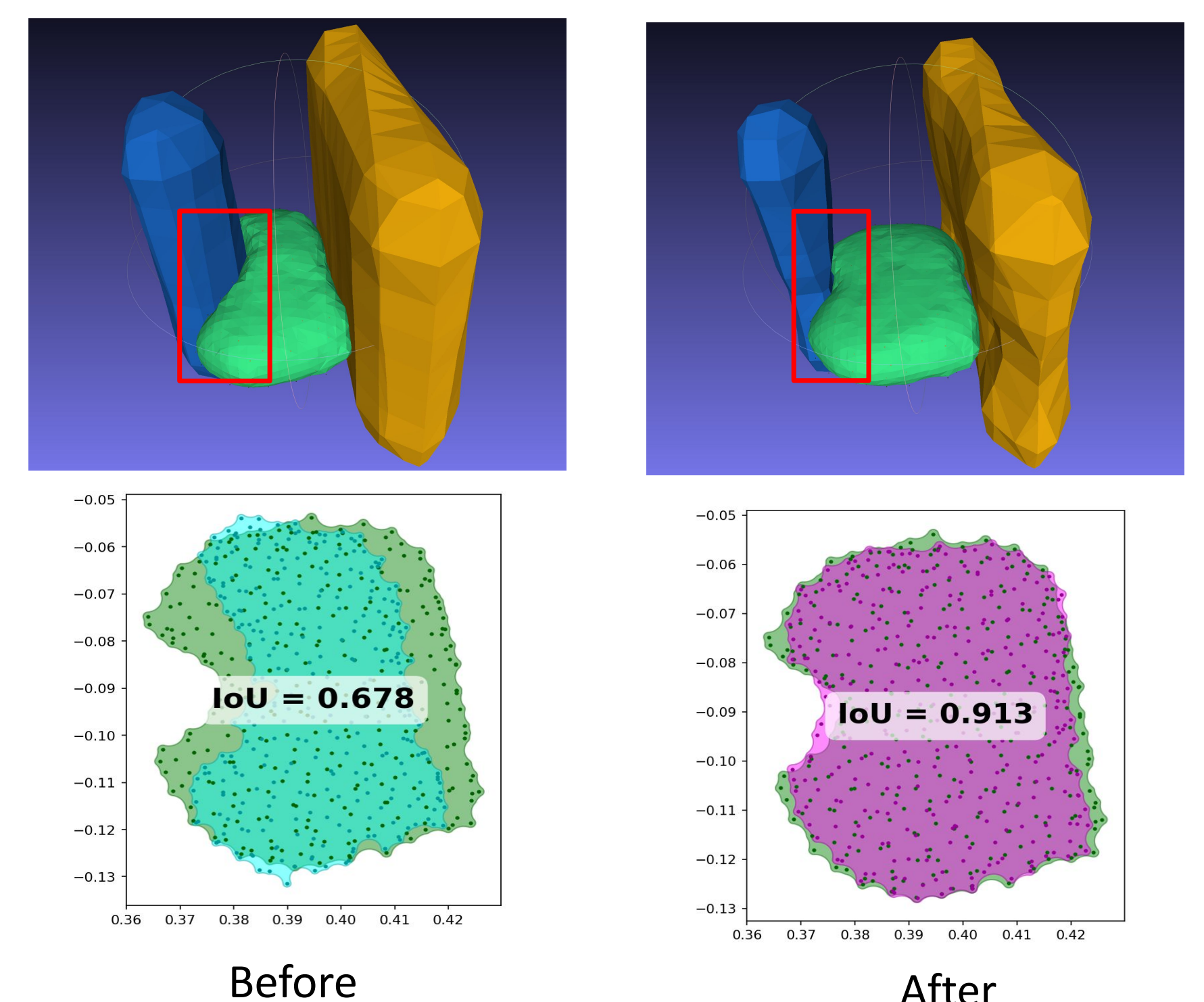
Optimize tool shape such that rollout matches the target goal shape!



Quantitative Evaluation: Inflate-then-Recover

Inflation %	Inflated / GT (CD %)	Optimized / GT (CD %)
25	110.9	101.2
50	122.0	101.2
100	145.9	103.5

Optimization



Conclusion

- Multi-tool training leads to generalization to unseen tool geometries.
- Improved shape reconstruction and fewer interpenetrations on real data.
- A differentiable dynamics model enabling high-dimensional tool-shape design

References

- Shi, H., Xu, H., Clarke, S., Li, Y. and Wu, J., 2023. Robocook: Long-horizon elasto-plastic object manipulation with diverse tools, CoRL 2023.
- Peng, S., Jiang, C., Liao, Y., Niemeyer, M., Pollefeys, M. and Geiger, A., 2021. Shape as points: A differentiable poisson solver, NeurIPS 2021.